

Appendix C - South-west Corner Marine Park (western arm) Data Analysis Report

South-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A.

Benthic ecosystem extent

Table C 1.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Benthic samples are pooled across survey years, so survey year was not offered as a candidate predictor.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Macroalgae	detrended+aspect+Z	0	0.707	0.63419	6.95
	detrended+roughness+aspect+Z	1.762	0.293	0.63419	7.81
Sand	detrended+roughness+as- pect+Z	0	1	0.50357	8.84
	detrended+roughness+as- pect+Z	0	0.997	0.38707	8.10
Rock	detrended+roughness+as- pect+Z	0	1	0.18945	8.49
	detrended+roughness+as- pect+Z	0	1	0.39016	8.96
Sessile invertebrates	detrended+roughness+as- pect+Z	0	1	0.39016	8.96
	detrended+roughness+as- pect+Z	0	1	0.45613	6.97
Reef	detrended+roughness+as- pect+Z	0	1	0.45613	6.97

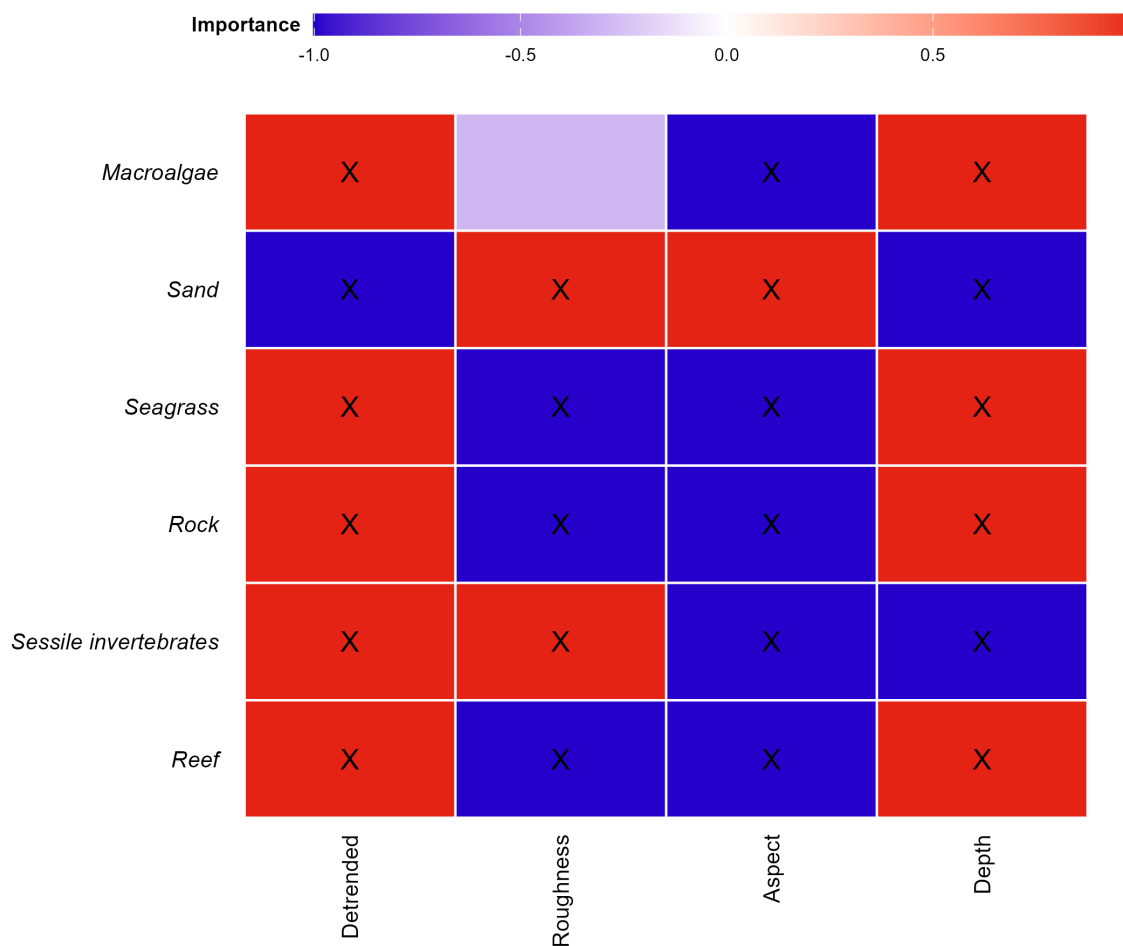


Figure C 1.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

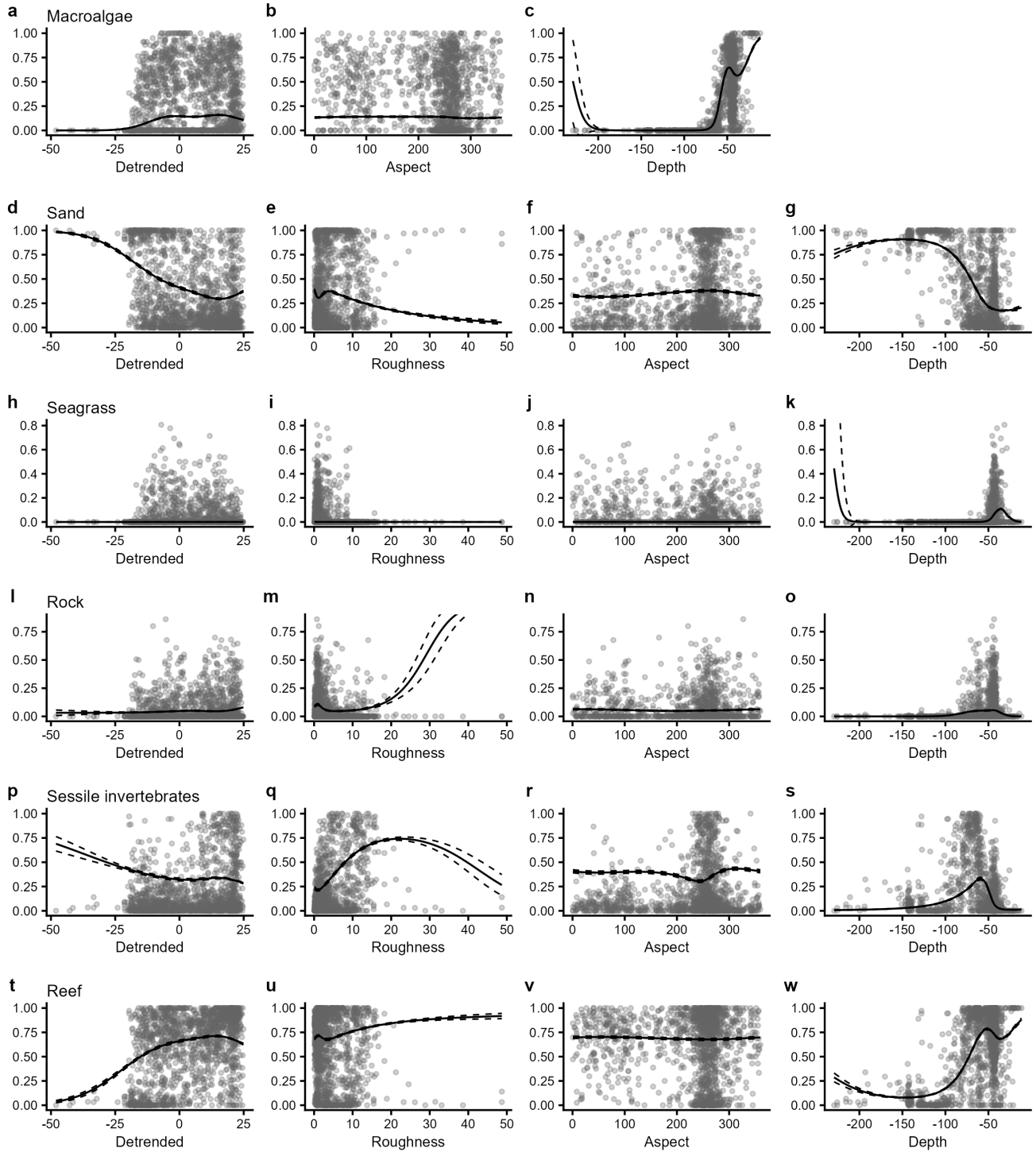


Figure C 1.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Benthic samples from 2020, 2021, 2023, 2024 and 2025 are pooled, because the campaigns sample different areas and a survey-year term would read as a spatial difference rather than change through time. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data.

Demersal fish assemblage

Table C 2.1: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	ΔAICc	ωAICc	AICc	R^2	EDF
Species richness	detrended+roughness+aspect+Z+year	0	0.898	3509.32	0.39035	10.17
Total abundance	detrended+roughness+aspect+Z+year	0	0.411	6671.44	0.36931	10.70
	detrended+aspect+Z+year	1.134	0.233	6672.57	0.36651	9.79
Large Reef Fish Index*	roughness+reef+year+status	0	0.456	11474.82	0.14542	7.88
	roughness+aspect+reef+year+status	0.052	0.444	11474.87	0.14547	8.88
	roughness+aspect+reef+year+status	0	0.252	1702.54	0.07864	8.19
Reef Fish Thermal Index	roughness+reef+year+status	0.001	0.252	1702.54	0.07864	7.20
	roughness+reef+year	0.765	0.172	1703.30	0.07375	6.00
	roughness+aspect+reef+year	0.766	0.172	1703.30	0.07375	7.00

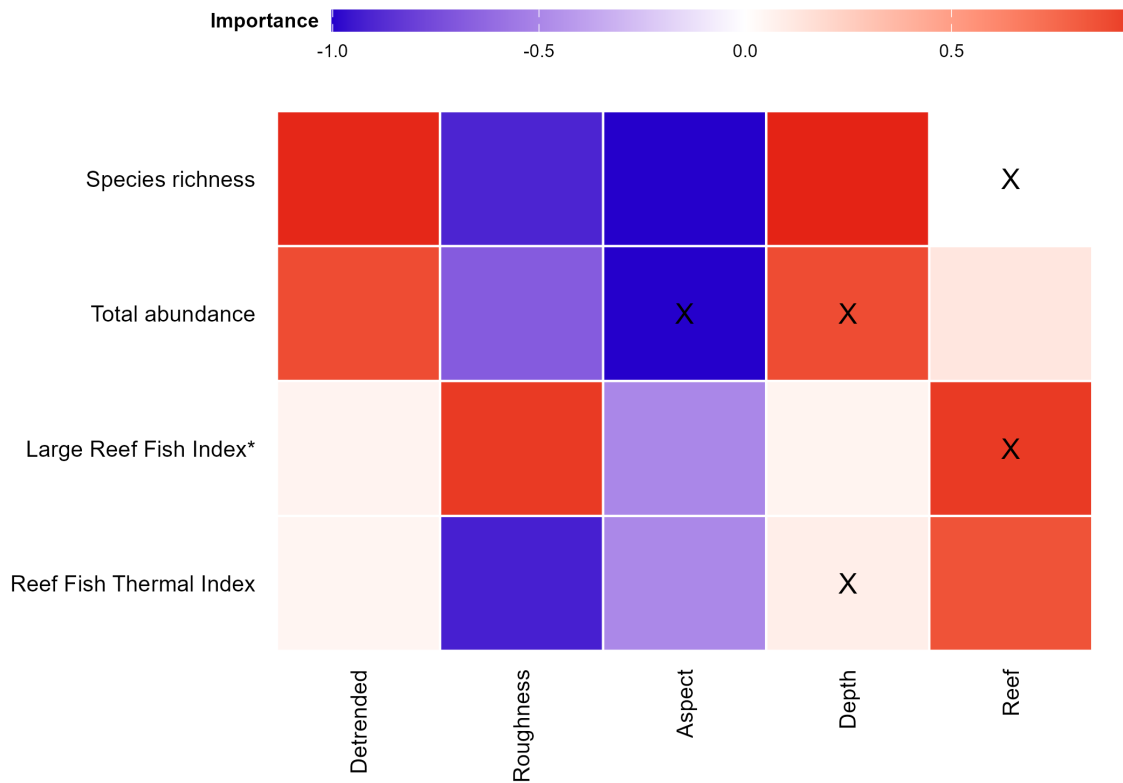


Figure C 2.1: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

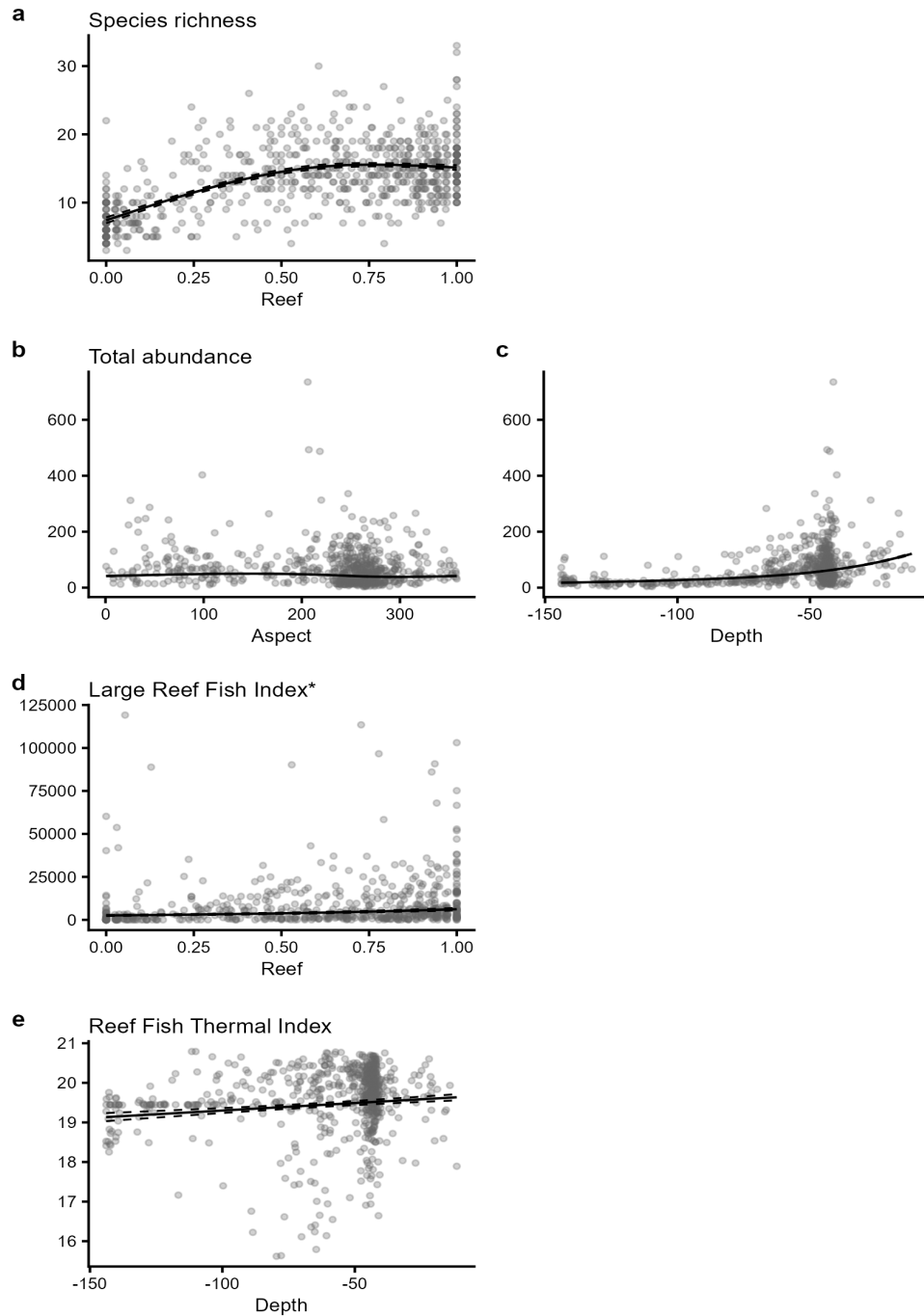


Figure C 2.2: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2020 and zone status held at areas open to fishing. These terms are held only where model selection retained them: year was not retained for Species richness and zone status was not retained for Total abundance and Reef Fish Thermal Index. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.